

SACHIN KUMAR

AI/ML Engineer & Data Scientist | Production LLMs, Multi-Agent Systems & Predictive Modeling

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PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with 1.8+ years of experience building and deploying production-grade LLM applications, multi-agent systems, RAG pipelines, predictive analytics, and autonomous infrastructure remediation swarms. Holds an M.Sc. in Business Statistics with expertise combining mathematical statistics, deep learning, and advanced Python/SQL to drive measurable enterprise outcomes, including 91% incident MTTR reduction and 99% latency optimization at Tata Consultancy Services.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Dec 2023 – Present | Chennai, India

Data Scientist & AI Automation Systems Engineer

- ISAcS Zero-Touch Incident Auto-Resolution Swarm: Architected an autonomous multi-agent system resolving enterprise ServiceNow ITSM incidents without human intervention, reducing Mean Time to Resolution (MTTR) from 4 hours to 20 minutes (91% reduction).
- High-Throughput Server Latency Optimizer: Engineered an asynchronous distributed event processing pipeline cutting peak server execution latency from 4 minutes to under 3 seconds (99% reduction).
- Anomaly Detection & Incident Correlation: Implemented density-based clustering algorithms across multi-system server telemetry logs, accelerating incident triage speed by 45%.
- Custom Model Context Protocol (MCP) Database Server: Built a natural-language-to-SQL MCP agentic tool for secure, read-replicating enterprise databases, increasing analytics query throughput by +36%.
- Production Enterprise RAG Pipeline: Deployed hybrid dense-sparse vector retrieval with cross-encoder re-ranking for technical knowledge bases, achieving 92% retrieval accuracy and zero hallucinated runbooks.

KEY PROJECTS & HONORS

Google Cloud Technical Debate — First Place Winner

2024

- Awarded 1st place in regional technical debate evaluating multi-agent orchestration patterns, cloud scalability, and enterprise RAG reliability.

Autonomous Multi-Agent Workflow Engine

- Designed specialized supervisor and worker agents with dynamic state machines and automated retry backoffs for IT operations.

TECHNICAL SKILLS

Generative AI & LLMs:	LangChain, LangGraph, Multi-Agent Systems, RAG, MCP (Model Context Protocol), Prompt Engineering, Vector Databases
Machine Learning & Stats:	Predictive Modeling, Time-Series Forecasting (ARIMA, Prophet), Anomaly Detection, Clustering, Hypothesis Testing
Languages & Frameworks:	Python (NumPy, Pandas, SciPy, FastAPI, Flask), SQL, TypeScript/Node.js, Bash scripting
Databases & Cloud:	PostgreSQL, MySQL, Redis, Google Cloud Platform (GCP), Docker, Linux, CI/CD, Git, ServiceNow API

EDUCATION

Master of Science (M.Sc.) in Business Statistics

University of Madras | First Class with Distinction

Bachelor of Science (B.Sc.) in Statistics

University of Madras | First Class